File 66: Master Geomorphic Restoration Workflow & Trout Spawning Ecology Guide

The 10-Step Regulatory & Fluvial Lifecycle for Wild Brown Trout Spawning Habitats

> ***Showcase Reference and Workflow Standard**: This guide provides Co-Founder Hunter Morris, Trout Unlimited, and high-value B2B partners with the absolute 10-step strategic and operational workflow for executing geomorphic stream restorations. It maps every phase from early landowner joint-venture underwriting to final 7-year regulatory credit releases. Crucially, it leads with the biological and physical chemistry standards required to restore wild **Brown Trout (`Salmo trutta`)** habitats, outlining how geomorphic channel modifications naturally regulate water temperature, maximize dissolved oxygen, and scour away suffocating silts from gravel spawning beds (redds).*

I. The 10-Step Stream Restoration & Mitigation Workflow

Every Blue Ridge Stream Restoration project operates under a strict geomorphic and regulatory sequence, ensuring maximum ecological benefit for trout waters and high-margin credit yields. Use this 10-step workflow to systematically replicate our stream water mitigation success on new streams:

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                   THE 10-STEP STREAM RESTORATION FLOW

[ 1. SOURCING ] ■■■> [ 2. GEOMORPHIC SURVEY ] ■■■> [ 3. 1D/2D HYDRAULIC MODEL ]
  Landowner JV,        Baseline thalweg,              LiDAR DEM meshing,
  HUC-8 audit          Wolman pebble count            HEC-RAS shear curves
                                                                ■
                                                                ▼
[ 6. CONSTRUCT ] <■■■ [ 5. EPD VARIANCE ] <■■■■■■ [ 4. USACE PCN PERMIT ]
  Pump-around,           25-ft protective buffer       Section 404 approval,
  NCD structure placements  sediment controls          RIBITS initial release
       ■
       ▼
[ 7. BIOENG ] ■■■■■> [ 8. AS-BUILT REPORT ] ■■■■■> [ 9. 7-YR MONITOR & RELEASE ]
  Live staking,        Post-construction monument     Annual EPT sampling,
  riparian canopy planting  cross-sections             redd count, credit payouts
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Step 1: Watershed Sourcing & Landowner Joint Venture (JV)

Identify degraded Agricultural/Forestry tracts (>= 50 acres) in high-priority HUC-8 basins exhibiting low competitor credit supply on RIBITS. Approach landowners using geomorphic legacy storytelling. Execute a 70/30 or 60/40 **Mitigation Joint Venture Agreement**, ensuring zero upfront cost to the landowner in exchange for a perpetual 50-foot riparian buffer conservation easement.

Step 2: Preliminary Fluvial Geomorphic Field Survey (Pre-Work Baseline)

Establish permanent monument cross-sections and longitudinal profiles using RTK GNSS and laser levels. Execute Wolman pebble counts to calculate the median substrate diameter (D■■). Conduct Bank Erosion Hazard Index (BEHI) and Near Bank Shear Stress (NBS) surveys to quantify annual lateral bank sediment loading. Install continuous HOBO temperature loggers to record summer thermal baselines.

Step 3: 1D & 2D Hydraulic & Sediment Transport Modeling

Import high-resolution DJI drone photogrammetry and LiDAR DEMs into HEC-RAS. Run steady and unsteady flow simulations to model boundary shear stresses under 1-year, 2-year, and 100-year storm events. Ensure the geomorphic design narrows the channel to keep sediment moving during high-flow events while maintaining shear stresses within stable thresholds (1.46 N/m² to 21.85 N/m²) to prevent bed scour.

Step 4: USACE Section 404 Nationwide Permit 27 (NWP 27) Submission

Compile and submit the Pre-Construction Notification (PCN) package to the USACE Savannah District. Detail the geomorphic Natural Channel Design (NCD) structure layouts (rock J-hooks, cross-vanes, step-pools) and calculate the projected stream credit yield. Execution of the conservation easement triggers the **Initial 15% Credit Release** on the federal RIBITS ledger.

Step 5: Georgia DNR EPD Trout Stream Buffer Variance Application

Submit a trout stream buffer variance application to the Georgia Environmental Protection Division (EPD) to authorize heavy machinery access within the state-protected 25-foot riparian buffer. Incorporate a robust **Fluvial Erosion & Sedimentation Control Plan**, including high-velocity pump-around bypass piping, Turbidity Barriers, and Coir Fiber Matting.

Step 6: Heavy Machinery Mobilization & Instream Excavation

Mobilize 13-to-15 ton excavators utilizing vegetable-based biodegradable hydraulic fluid to protect trout water chemistry. Construct temporary sandbag cofferdams upstream and pump streamflow around active 100-foot zones. Grade bank slopes back to a stable 3:1 profile, excavate deep thermal pool thalwegs (4.5 ft depth), and install geomorphic rock and log structures.

Step 7: Bioengineering & Riparian Shading Canopy Installation

Stabilize newly graded outer banks immediately using double-net coir matting (700g/sqm) secured with oak stakes. Insert dormant **Black Willow and Silky Dogwood Live Stakes** at 2-foot intervals. Plant high-density, bare-root native trees (Eastern Hemlock, River Birch, Rosebay Rhododendron) across the 50-foot riparian zone to lock the soil matrix and establish thermal shading.

Step 8: Post-Restoration Construction Verification & As-Built Report

Rerun longitudinal and lateral monument cross-section surveys immediately following construction to establish the baseline "As-Built" geomorphic geometry. Submit the **As-Built Monitoring Report** to the USACE Interagency Review Team (IRT) to trigger the **Second 10% Credit Release** on RIBITS.

Step 9: 7-Year Annual Ecological & Geomorphic Monitoring

Field crews monitor the restoration reach annually for 7 years, collecting geomorphic stability cross-sections, continuous temperature data, substrate pebble distribution data, and biological assessments. Benthic macroinvertebrate sampling must prove an increasing trend in sensitive aquatic taxa (mayflies, stoneflies, caddisflies), and annual spawning surveys must map active trout nests.

Step 10: Phased Credit Releases & Financial Asset Liquidation

Achieving annual geomorphic and biological performance standards triggers successive credit releases on RIBITS (typically 10% to 15% annually). Liquidate stream mitigation credits to commercial buyers, exurban developers, and corporate data centers at current market rates (110 to 140 USD per credit), distributing the net cash flows to partners under the Joint Venture ledger.

II. Wild Brown Trout Spawning Ecology & Fluvial Mechanics

A successful stream mitigation project must lead with the biological and physical chemistry requirements of its target species. For our North Georgia showcase projects, the ultimate ecological indicator is the wild **Brown Trout (`Salmo trutta`)**.

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                   BROWN TROUT HABITAT RESTORATION PHYSICS

[ RIPARIAN CANOPY SHADE ] ■■■■■■■■■■■■■■■■■■■■■> [ COLD WATER TEMP (<65°F) ]
  Attenuates solar heat flux                    Maximizes dissolved oxygen
                                                         
                                                         ■
                                                         ▼
[ GRAVEL HYDRO-DYNAMICS ] <■■■■■■■■■■■■■■■■■■■■■ [ GEOMORPHIC FLUSHING ]
  Stokes' Law silt transport,                   Rock J-Hooks & Cross-Vanes
  keeps egg gravels clean (<5% silt)            concentrate hydraulic shear
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1. Thermal Physics & Heat Attenuation Models

Brown trout are coldwater stenotherms, exhibiting strict physiological limits. Water temperature governs their metabolic rates, spawning cues, and egg survival:

- **Optimal Spawning Range**: 48°F - 54°F (9.0°C - 12.2°C) in late autumn (October to December).
- **Optimal Growth Range**: 50°F - 60°F (10.0°C - 15.6°C).
- **Lethal Thermal Stress Threshold**: Exceeding 68°F - 72°F (20.0°C - 22.2°C). Warm water causes physiological stress, suppresses immune response (triggering *Saprolegnia* fungal infections), and causes mortality.

Riparian Canopy Solar Energy Attenuation Model
To lower summer temperatures, we plant dense native riparian vegetation. The reduction of solar radiation heat flux (I) passing through the vegetative canopy is modeled using the **Beer-Lambert Law**:
I = I_0 \cdot e^{-k \cdot \text{LAI}}

- *Where*:
- I■ is the incident solar radiation heat flux (W/m²) reaching the upper canopy limit.
- I is the remaining solar radiation heat flux reaching the stream surface.
- k is the canopy extinction coefficient (dimensionless, typically 0.5 - 0.7 for dense deciduous-coniferous cover).

- LAI is the Leaf Area Index (m^2 leaf area per m^2 ground area).
- *Restoration Design Goal*: To achieve a **96% solar radiation attenuation** (I/I■ = 0.04), our bioengineering team must restore a mature riparian buffer consisting of coniferous Eastern Hemlock and understory broadleaf Rosebay Rhododendron, achieving a design target **LAI >= 5.8**.

2. Dissolved Oxygen Physical Chemistry & van 't Hoff Equations

Trout absorb oxygen through their gills via passive diffusion, requiring high concentrations of Dissolved Oxygen (DO) in both the water column and the spawning gravel beds.
- **Water Column DO Standard**: >= 7.0 mg/L (optimal >= 9.0 mg/L).
- **Intragravel (Substrate) DO Standard**: >= 6.0 mg/L within spawning beds.
- **Trout Redds (Nests)**: Female brown trout excavate nests (redds) in gravel riffles, deposit eggs, and cover them. The survival of these buried eggs depends on continuous hyporheic downwelling flow carrying oxygenated water through the gravel pores.

Temperature-Oxygen Solubility Model
The saturation concentration of dissolved oxygen (Cs, mg/L) in water is inversely proportional to water temperature. We model this physical chemistry using the modified **van 't Hoff equation**:
$$C_s = C_0 \cdot e^{-\Delta H_{\text{sol}} \cdot \left(\frac{1}{T} - \frac{1}{T_0}\right)}$$
- *Where*:
- C■ is the baseline dissolved oxygen solubility concentration (14.6 mg/L) at reference temperature T■ = 273.15 K (0°C).
- ΔH_{sol} is the molar enthalpy of oxygen dissolution in water (approximately -12.4 kJ/mol).
- T is the actual water temperature (in Kelvin).
- *The Ecological Impact*: As water warms from 50°F (283.15 K) to 71.8°F (295.26 K), oxygen solubility drops by over **26%**, reducing available oxygen from 11.3 mg/L to a hypoxic 8.3 mg/L at saturation, suffocating developing eggs. Keeping streams cold is physically necessary to keep oxygen levels high.

3. Geomorphic Gravel Flushing & Stokes' Law Sediment Transport

The most severe threat to brown trout spawning success is **sedimentation (redd clogging)**. When upstream bank erosion is high, fine sand, silt, and clay particles (<2mm) settle onto spawning gravels. This causes:
1. **Redd Cementation**: Fine sediments seal the gravel interstitial spaces, blocking hyporheic downwelling flow and reducing oxygen supply to the eggs.
2. **Toxic Accumulation**: Clogged gravels trap metabolic wastes (carbon dioxide, ammonia) produced by the developing embryos, causing toxic mortality.

Stokes' Law Sediment Settling Velocity
To model how silt settles and how geomorphic restoration structures prevent it, we calculate the terminal settling velocity (vs) of sediment particles using **Stokes' Law**:
$$v_s = \frac{g \cdot (\rho_p - \rho_f) \cdot d^2}{18 \cdot \mu}$$
- *Where*:
- vs is the settling velocity of the sediment particle (m/s).
- g is the acceleration due to gravity (9.81 m/s^2).
- ρ_p is the density of the sediment particle (2,650 kg/m^3 for quartz sand/silt).

- ρ_f is the density of water (1,000 kg/m^3).
- d is the sediment particle diameter (m).
- μ is the dynamic viscosity of water (1.002 x 10■^{3} Pa·s at 20°C).
- *Restoration Design Goal*: To prevent fine silt (d = 0.05 mm) from settling onto spawning gravels, the local shear velocity and upward turbulent fluid forces must exceed the settling velocity (vs = 0.0022 m/s).
- *Fluvial Solution*: We place **Rock J-Hook Vanes** and **Log Step-Pools** that contract the channel bankfull width by **50%**, concentrating streamflow down the thalweg. This geomorphic constriction locally elevates shear velocity during high-flow events, keeping fine silts in suspension and scouring the spawning gravel beds (D■■ = 22mm) clean.

III. Summary Checklist for Sourcing a New Spawning Project

When evaluating a new stream restoration candidate for partnership, verify these baseline biological and geomorphic conditions:

- [ ] **Thermal Baseline**: Summer peak temperatures must be capable of being cooled to <65°F via riparian canopy restoration.
- [ ] **Substrate Sizing**: Channel must exhibit gravel-to-cobble bed material (D■■ median target sizing of 15mm - 40mm for brown trout spawning).
- [ ] **Erosion Sourcing**: Active bank erosion (BEHI: High or Extreme) must be present upstream to generate a significant sediment-reduction credit yield under USACE math.
- [ ] **Hydraulic Grade**: Reach slope must support step-pool or riffle-pool sequences (0.5% - 2.0% gradient) to drive natural hyporheic gravel flushing.
- [ ] **Landowner JV**: Settle a mutually beneficial 70/30 joint-venture split with UETA-compliant digital signature integrations.

Developed by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin.